

Civics Group		Name (use BLOCK LETTERS)
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H2

Centre number / Index Number	
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**ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIMS**

H2 BIOLOGY
9744/4
Paper 4: Practical Exam

Friday

24th August 2018

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use a HB pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

IMPORTANT INFORMATION TO CANDIDATES:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use them. Please answer **QUESTION 2** within this time frame.

Candidates with no access to microscope at the start of the paper will be given access **1h 15min after the start of the paper**. You may proceed with **QUESTION 1** first.

Shift
Lab

For Examiner's Use	
Section A	
1	/ 20
2	/20
3	/15
Total	/55

QUESTION 1

Investigation of glucose concentration in fruit juice.

You are advised to:

- Read through the entire question and prepare a table to record your results in **(b)(ii)**;
- Prepare a boiling water bath;

before starting the investigation.

You are required to estimate the concentration of reducing sugar in a fruit juice, labelled **F1**, by comparison with that in a range of glucose solutions.

You are provided with a 0.8% solution of glucose, labelled **F2**, Benedict's solution, distilled water and five test-tubes.

(a) Carry out the Benedict's test on fruit juice **F1**. Describe the procedure. Record your results in your table in **(b)(ii)**.

.....
[1]

(b) You are now going to test a range of glucose solutions that you will prepare yourself using **F2** and distilled water.

Carefully follow the instructions below. You should present and record your observations and data in the space provided. You will need to:

- read through the instructions carefully
- prepare the space on the next page so that it is ready for you to record the readings
- prepare a range of glucose of varying concentrations by **serial dilution**
- carry out the tests so that you can compare your results with the result for the fruit juice **F1**.
- rinse your syringes if necessary.

(i) You will now perform **serial dilution** using **F2** (0.80% glucose) as stock solution according to the steps below.

You will use **F2** (0.80% stock solution) to prepare 8.0 cm³ of glucose solution of **half** the concentration i.e. 0.40%.

Using this newly prepared 0.40% solution, prepare another 8.0 cm³ solution of **half** the concentration i.e. 0.20%.

Continue using this method until you have a total of 5 glucose solutions of varying concentrations.

Complete the table below.

.....[2]

concentration of glucose / %	0.80 (F2)				
volume of the glucose solution to be diluted / cm ³					
volume of distilled water / cm ³					

(ii) Using a **table**, record the results for Benedict's test of F1 and the range of glucose standards in the space below.

.....[3]

(iii) Estimate the concentration of reducing sugar in the fruit juice **F1**. You may use a range for your estimation or the nearest glucose standard concentration.

..... [1]

(c) The volume of reactants can affect the results that you obtain.

(i) State how you controlled this variable in your investigation.

.....
.....[1]

(ii) Identify two other significant sources of error in this experiment.

1
.....
2
.....[2]

(d) Suggest how the student could improve this experiment.

.....
.....
.....
.....
.....
.....[3]

(e) A student used another carbohydrate, starch, to investigate the effect of pH on the activity of the enzyme amylase.

The data in Table 1.1 were obtained.

Table 1.1

pH	time taken for complete hydrolysis / min			mean time / min
	first run	second run	third run	
5	9	10	8	9.0
6	7	6	8	7.0
7	3	4	3	3.3
8	4	5	6	5.0
9	10	9	11	10.0

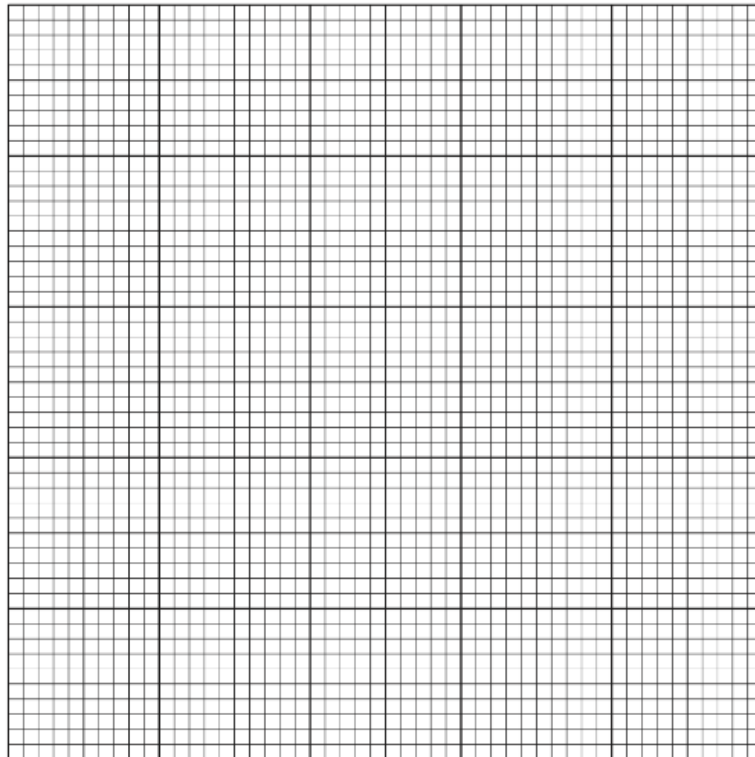
(i) When the student first performed this investigation, the time taken for complete hydrolysis at pH 7 was 17 minutes.

Explain why the student discarded this result and repeated the experiment with freshly made solutions.

.....
[1]

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(ii) Plot a graph to show the effect of pH on the time taken for complete hydrolysis of starch by amylase.



[3]

(f) The student's hypothesis was:

- the activity of the enzyme would increase with increasing pH.

Discuss the student's hypothesis in relation to the results obtained.

.....

.....

.....

.....

.....

.....[3]

[Total : 20]

QUESTION 2

You will need access to a microscope and stage micrometer for this question.

Betalains are water-soluble red pigments found in some fungi and flowering plants, including rhubarb and beetroot.

The red colour of the rhubarb stem is a result of betalains, present in the cytoplasm of the rhubarb cells. This gives the cytoplasm a reddish colour, enabling the cell membrane to be distinguished.

You are required to investigate the effect of a solution, **labelled K**, on the epidermal cells of the rhubarb.

You are provided with a petri dish containing two segments of the rhubarb stem.

- Use a pair of forceps to carefully peel a piece of the epidermis (coloured) from one of the rhubarb stems.
- With the scalpel, cut a small piece (e.g. 0.5cm X 0.5cm) from this epidermal tissue.
- Mount the square on a microscope slide in **distilled water** and cover it with a cover slip.

You are also provided with an eyepiece graticule that has been fitted to the eyepiece of your microscope and a stage micrometer.

The 1cm stage micrometer is divided into 100 divisions.

(a) Observe your slide under the low-power and then high-power objective lens (40X) of your microscope.

(i) Use the space below to make a high-power drawing of **3** adjoining epidermal cells. Label your drawing.

.....[5]

(ii) Count the number of divisions of the eyepiece graticule across any 1 of the 3 cells using the 40X objective lens.

Number of divisions

Remove the slide and replace it with the stage micrometer. Using the same magnification, adjust the focus until you can see the eyepiece graticule on top of the stage micrometer.

- (iii)** Count the number of eyepiece graticule divisions that match an exact number of stage micrometer divisions.

Number of eyepiece graticule divisions

Number of stage micrometer divisions

Use this information to calculate the actual diameter of the cell.
Show your working.

Actual diameter of cell [2]

- (iv)** Suggest how an error in measuring the lengths could occur.

.....

.....[1]

- (b)** Fig. 2.1 shows a photomicrograph of rhubarb cells after being immersed in **Solution K** for a few minutes.

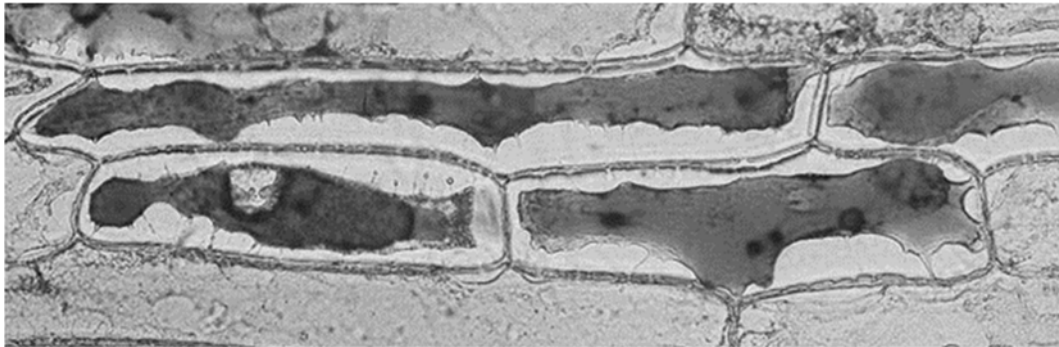


Fig. 2.1

- (i) Prepare the space below so that it is suitable for you to record the observable differences between the rhubarb cells you have seen in **2(a)** and the cells in **Fig. 2.1**.

Record these differences in the space you have prepared.

.....[2]

- (ii)** Explain the observation of the cells in **Fig. 2.1** and suggest a property of **Solution K**.

[4]

- (iii)** With atmospheric temperature set to increase due to climate change, predict how the constituents of the plasma membranes of plants will change.

Explain why.

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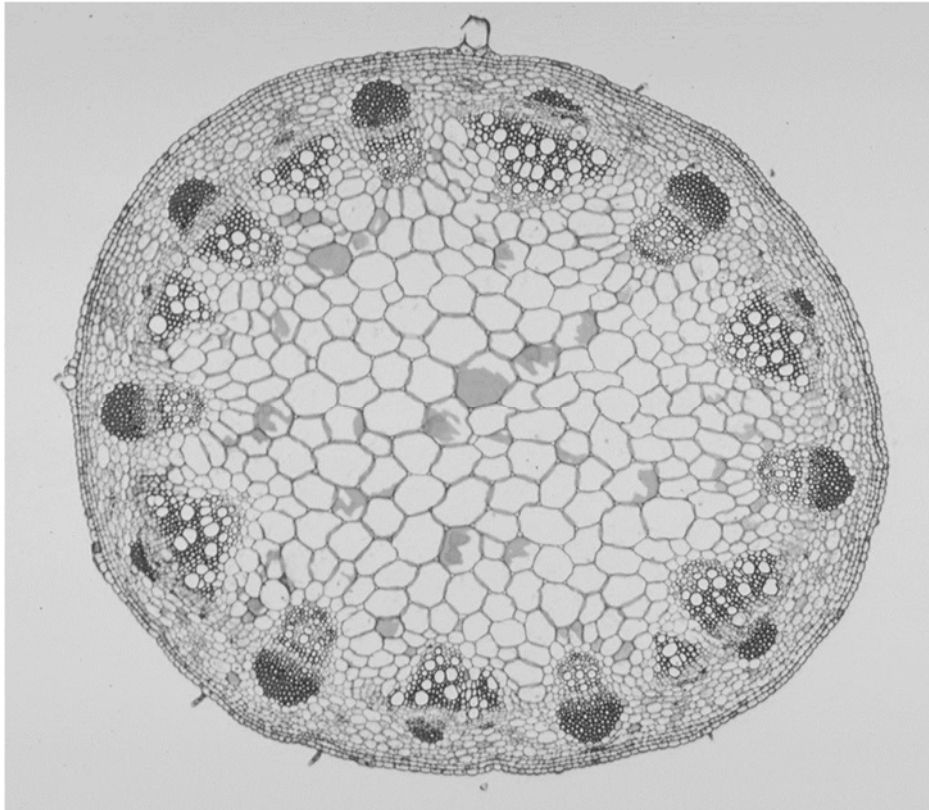
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.....

.....[3]

(c) **Fig. 2.2** is a photomicrograph of a stained transverse section through an organ of a different type of plant.

You are not expected to be familiar with this specimen.



(Adapted from www.carolina.com)

Fig. 2.2

Make a low-power plan drawing of **Fig. 2.2**.

.....[3]

[TOTAL : 20]

QUESTION 3: PLANNING QUESTION

Effect of alcohol on membrane permeability of beetroot tissue.

Beetroot cells store molecules of betalain in their vacuoles. The membrane surrounding vacuoles in plant cells is the tonoplast, which has a similar structure to other cell membranes. Various factors such as alcohol influence the permeability of membranes.

Using this information and your own knowledge, design an experiment to determine the effect of different concentrations of alcohol on the membrane permeability of beetroot cells.

The following equipment may be used in your plan or not as you wish. You should **not** use any additional reagents/equipment.

- beetroot root tissue
- distilled water
- a supply of hot water
- Clingwrap
- 100% alcohol containing a mixture of ethanol and methanol
- thermometer
- thermostatically controlled water bath
- colourimeter & cuvettes
- stopwatch
- core borer
- ruler
- scalpel
- syringes
- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 15]

